Home plan: Sprouts

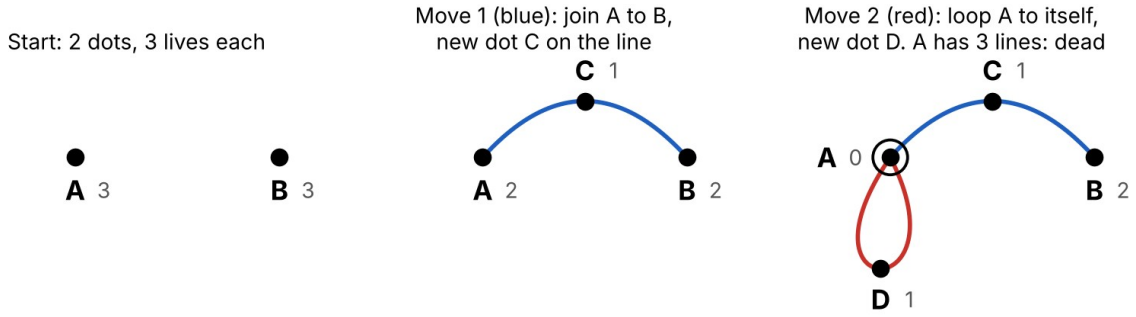
Sunday, November 8, 2026, at the kitchen table: James with Jay, Rose and perhaps a third child. **Untried:** nobody has played this plan yet, and every time is a guess.

What you're doing

You play a pencil game called Sprouts. Draw two or three dots. Players take turns drawing a line from a dot to a dot (or back to the same dot) and putting a new dot on that line. Lines may not cross, and a dot with three lines leaving it is dead. Whoever cannot move loses. Rose has a one-dot game she always wins by going second, and she is the dead-dot spotter. Jay hunts for the longest and shortest games and finds out why a game with n dots can never last more than 3n − 1 moves, by giving every dot three "lives". You need paper and two pens of different colours. Play 20 to 30 minutes and stop whenever.

The rules

- Start with a few dots, a hand-width apart.
- On your turn, draw a line from one dot to another, or a loop from a dot back to itself. Then put a new dot anywhere along your line (not on its ends).
- A line may bend and go round other dots and lines, but may not cross itself or another line, and may not pass through a dot.
- A dot can have at most 3 lines leaving it. Count a loop as 2 at its dot, and a new dot as 2 already (one each way along the line it sits on), so it can take one more. So a loop needs a dot with room for 2 more.
- If you cannot move, you lose. Said the other way: whoever makes the last move wins.
- Each player draws in their own colour, so you can see afterwards who did what.



Grey numbers: lives left (3 minus lines leaving the dot). Ring: a dead dot.

Lives. Think of every dot as having 3 lives, and every line leaving it as using one. A dot with 0 lives is dead: draw a ring round it so nobody tries to use it. A new dot has 1 life.

Counting moves. No tally is needed: every move adds one dot, so the number of moves is the number of new dots.

The mathematics, for you

1. **Every game ends, within 3n − 1 moves (n = starting dots).** *Proof.* The dots start with 3n lives. A move uses 2 lives (its two line ends) and adds a dot with 1 life, so the total drops by exactly 1 each move. After m moves there are 3n − m lives. The newest dot always has a life (nothing has been drawn since it was made), so the total never reaches 0, and m ≤ 3n − 1. A check you can use at the table: moves made + lives left = 3n, always. *Limits:* the count needs exactly 3 lives. With 4 lives a new dot keeps 2 and the total never drops, and the newest dot can loop for ever.
2. **Every game lasts at least 2n moves.** *Proof sketch, only for the keen.* At the end each live dot has exactly 1 life and shares no region with another live dot. Each live dot then has two dead neighbours (dots joined to it by a piece of line) that no other live dot touches; a dead dot has three pieces, and any two border one region, so it can't serve two live dots. (If a live dot sits on a loop, its two dead dots are the loop's dot and that dot's third neighbour.) So 3 × (live dots) ≤ all dots = n + m, and live dots = 3n −

m gives m ≥ 2n.

3. **What actually happens** (computer search in our checker). 1 dot: always exactly 2 moves, so the second player wins whatever anyone does. 2 dots: 4 or 5 moves; the second player wins with best play. 3 dots: 6, 7 or 8 moves; the first player wins with best play, and we know no short rule for how. For 1 to 3 dots, every length from 2n to 3n − 1 happens. Since moves = 3n − (dots alive at the end), the game is really a fight over how many dots survive: an odd number of moves means the first player made the last one. Cited, not rechecked: 4 and 5 dots are first-player wins and 6 a second-player win (Applegate, Jacobson and Sleator, 1991, by computer), and nobody has proved a pattern. Sources: Conway and Paterson invented Sprouts (1967); Gardner, *Mathematical Carnival*, chapter 1; Berlekamp, Conway and Guy, *Winning Ways*, chapter 17. Our checker, src/sprouts_check.py, verifies every other fact here.
4. **Your strategy going second on 2 dots** (dots A and B; the new dots are C, D, ...).
    - If they join A to B: **join A to B again.** That makes a closed circle of four dots, one life each. Whatever two of those dots they join next, **join the other two, on the other side of the circle.** The game is over after 4 moves, and you made the last one.
    - If they loop A (or B): **join that dot to the new dot on its loop, on the side away from the other dot.** The other dot is then on its own, it takes exactly two more moves, and you make the second.

What children might find. By experiment: games always end; the same start gives games of different lengths; one dot is always 2 moves. As conjectures: "2 dots never takes more than 5", "second wins with 1 dot". As explanations: lives, which settle the longest game for any number of dots. The shortest-game proof is harder; most children stop at "it seems to be at least 2n".

Materials and setup (2 minutes)

- Plain paper, one game per half sheet, and two pens of different colours per game. Rose may prefer a fat marker.
- Spread the starting dots a hand-width (about 5 cm) apart, so lines have room.
- Sit side by side so everyone sees the drawing the same way up.

The route

Each part stands alone. Stop after any of them and pick up anywhere next time.

1. Launch (3 minutes). Draw two dots. Play the two moves in the picture yourself, saying the rules as you go: "a line, then a new dot on it"; "A has three lines now, so I ring it: it's dead." Let Jay and Rose finish that game against each other. Say only whether a move is allowed. *Watch for* a forgotten new dot, a line through a dot, and a loop counted as one line instead of two.

2. Rose's part (5 to 10 minutes).

- *One dot, she goes second.* You loop the dot. She joins the dot to your new dot, and the game is over: she wins. It always goes this way, whatever anyone draws. Play it a few times, then ask "With one dot, would you rather go first or second?" Then she can challenge Jay to it, telling him "You go first."
- *Dead-dot spotter.* In Jay's games, she checks each dot by touching the lines leaving it ("one, two, three: dead") and rings dead dots. If anyone disagrees, count together.
- *Two dots, as a team.* She teams with Jay or with you; she picks where the new dot goes.

Rose's moment: she wins the one-dot game knowing she will before it starts, or catches a live dot everyone else thought was dead. *Stop* when she would rather draw.

3. Jay's part (10 to 20 minutes). Ask one question at a time and let him play to find out.

- *Longest and shortest.* "Playing together, not against each other: what's the longest 2-dot game you can make? The shortest?" (5 and 4.)
- *Why.* "Why can't you make 6? Could a game go on for ever?" Hint if asked: "Give every dot 3 lives. What does one move do to the total?" To find: the total drops by 1; the newest dot always keeps a life; so at most 6 − 1 = 5.
- *Predict.* "Before we play 3 dots: longest? shortest?" (8 and 6.) Then "10 dots?" (29 and 20.) Play a 3-dot game to check the longest.

- *Beat me on 2 dots.* He chooses who goes first; going second, use the strategy above. If he loses twice, ask "What did I do after your first move?"
- *Who wins? Count from the end.* After a few 2-dot games: "When one dot was left alive, how many moves did it take? When two were left?" (5 and 4: moves = 3n − dots alive.) Odd means the first player made the last move, so the fight is over how many dots survive.

Only if he is keen: "Why can't a 3-dot game end in 5 moves?" Hint if asked: "At the end, look at a dot that's still alive. What is next to it?" The proof is item 2 above; reaching "every live dot has its own dead neighbours" is plenty.

4. Closing questions (open).

- "On 3 dots, would you rather go first or second?" (First, by computer; nobody knows a simple way to say how.)
- "What if every dot had 4 lives instead of 3? Does a game still have to end?" (No: the newest dot can loop for ever.)

When to stop. On a high point: Rose's win, or Jay's reason why 5 is the most. Next time, start with "Beat me on 2 dots" or with 3 dots.

How it went

(Add a dated line after the weekend.)